

PURCHASE DESCRIPTION FOR THE ACQUISITION OF LARGE PLASTIC MEDIA BLAST BOOTH

1.0 SCOPE.

1.1 Scope. The content of this specification describes the minimum requirements for the procurement, delivery, installation, and commissioning of a new Plastic Media Blast (PMB) booth for the Fleet Readiness Center, Southwest (FRCSW) in Building 472 Blast Shop. The blast booth shall be designed for mechanical removal of corrosion and surface preparation and cleaning of various aircraft components, utilizing PMB technology. The scope of this procurement includes, but is not limited to, removal, design and specifications, installation, testing and commissioning, documentation, and training.

2.0 APPLICABLE DOCUMENTS. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, Contractor shall meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

2.1 Government Documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract. The latest editions of the documents at time of solicitation shall be used.

STANDARDS – FEDERAL

FED-STD-H28B: Screw Thread Standards for Federal Service

(Applications for copies should be addressed to Defense Logistics Agency (DLA) Document Services, Attn: Standardization Documents Office, 700 Robbins Avenue, Building 2, Philadelphia, PA 19111-5096. For electronic access, contractors can also visit the ASSIST database at: <https://assist.dla.mil>)

U.S Department of Labor, Occupational Safety and Health Administration (OSHA)

29 CFR 1910: General Industry, OSHA Safety and Health Standards (2017 Ed.)

(Application for copies should be addressed to the OSHA Publications Superintendent of Documents P.O. Box 37535 Washington, D.C. 20013-7535)

Environmental Protection Agency (EPA)

40 CFR Part 761: Polychlorinated Biphenyls (PCBs) Manufacturing, Processing, Distribution in Commerce, and Use Prohibitions (2019 Ed.)

(Application for copies should be addressed to the EPA Ariel Rios Building
1200 Pennsylvania Avenue, N.W. Washington, DC 20460)

DEPARTMENT OF DEFENSE

U.S. NAVY

OPNAVINST 3960.16B: OPNAVINST 3960.16A: Navy Test, Measurement, and Diagnostic Equipment (TMDE), Automatic Test Systems (ATS), and Metrology and Calibration (METCAL) (2017 Ed.)

OPNAV 5100.23H: Navy Safety and Occupational Health Program Manual (2020 Ed.)

(Application for copies of the OPNAVINST should be addressed to Office of the Chief of Naval Operations, The Pentagon, Washington, DC 20350-2000 or to Commander, Naval Safety Center 375 A Street Norfolk, VA 23511 or on the internet at <http://www.public.navy.mil/navsafecen>)

MILITARY STANDARDS

MIL-STD-130N w/Change 1: Identification Marking of U.S. Military Property (2012 Ed.) Item Unique Identification Marking (IUID)

(Application for copies should be addressed to Air Force Product Data Systems Modernization Standards (TMSS) Office, 554 ELSG/SBP, 4375 Chidlaw Road, Bldg. 262, Room S008, Wright-Patterson AFB, OH 45433-5006, Phone (937) 257-0870 or 0853 or <https://quicksearch.dla.mil/qsSearch.aspx>)

MIL-P-85891A: Military Specification, Plastic Media, For Removal of Organic Coatings (1 APR 1992)

MIL-STD-1504C: Department of Defense Standard Practice: Abrasive Blasting (01 MAR 2007) [NO S/S DOCUMENT]

(Applications for these documents should be addressed to the contracting officer 30 days after contract award)

2.2 Non-Government Publications, Specifications, Standards, and Handbooks. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the documents listed in the solicitation or contract shall be the current edition at time of solicitation.

American Society of Mechanical Engineers (ASME)

ASME BPVC SEC VIII-1: Boiler and Pressure Vessel Code (2019 Ed.)

ASME Y14.100: Engineering Drawing Practices (2017 Ed.)

(Applications for copies should be addressed to ASME International Three Park Avenue New York, NY 10016-5990 or www.asme.org)

International Organization for Standardization (ISO)

ISO 14877: Protective Clothing For Abrasive Blasting Operations Using Granular Abrasives (2002 Ed.)

ISO 4414: Pneumatic power-General rules and safety requirements for systems and their components (2010 Ed.)

(Application for copies should be addressed to the International Organization for Standardization (ISO) 1 rue de Varembé, Case Postale 56, Geneva, CH-2111 Switzerland)

National Electrical Manufacturers Association (NEMA)

NEMA 250: Enclosures for Electrical Equipment (1000 V Maximum) (2018 Ed.)

NEMA ICS 1: Industrial Control and Systems General Requirements (2000 Ed. 2015R)

NEMA ICS 6: Industrial Control and Systems: Enclosures (1993 Ed. 2016R)

NEMA MG 1: Motors and Generators (2018 Ed.)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 N 17th Street, Rosslyn, VA 22209)

National Fire Protection Association (NFPA)

NFPA 13: Standard for the Installation of Sprinkler Systems (2019 Ed.)

NFPA 68: Standard on Explosion Protection by Deflagration Venting (2018 Ed.)

NFPA 70: National Electrical Code (2020 Ed.)

NFPA 79: Electrical Standard for Industrial Machinery (2018 Ed.)

(Applications for copies should be addressed to the National Fire Protection Association, ATTN: Support Services, 11 Tracy Drive, Avon, MA 02322)

National Institute of Standards and Technology (NIST)

NIST SP-250 Calibration Services Users Guide (1998 Ed.)

(Application for copies should be addressed to:
<http://nvlpubs.nist.gov/nistpubs/ir/2013/NIST.IR.7298r2.pdf>, and
http://www.nist.gov/calibrations/sp250_series.cfm)

2.3 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulation unless a specific exemption has been obtained.

3.0 REQUIREMENTS.

3.1 Design. The item offered shall be new and a current commercially available model. A current model is defined as the manufacturer's currently produced model which, on the date this solicitation is issued, has been designed, engineered, sold or is being offered for sale through advertisements or manufacturer's published catalogue or brochures. Products such as a prototype unit, pre-production model, or experimental unit do not qualify as meeting the requirements specified herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measuring and Indicating Device Graduations. All measuring and indicating devices on the machine shall be graduated using inch and pound units of measurement.

3.1.2 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's workstation(s).

3.1.3 Safety and Health Requirements. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and OPNAVINST 5100.23H.

3.1.4 Audible Noise Level. Audible noise emitted by the equipment shall not exceed 84 decibels (dB), measured on the "A" weighted scale of a standard Type II sound level meter, at the operator's work position or any point at a distance of three (3) feet from the equipment. Noise generated by the work piece shall be excluded in determining compliance of the equipment with the 84 dB requirements.

3.1.5 Mercury Restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture with the exception of fluorescent lighting where required.

3.1.6 Asbestos Restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.1.7 Lead. Lead and materials containing lead shall not be used on or in the machine with the exception to radiation cladding and lead solder in circuits.

3.1.8 Use of Polychlorinated Biphenyl (PCB). The use of polychlorinated biphenyl (PCB) on or in the equipment is prohibited.

3.1.9 Environmental Protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.1.10 Lubrication. Means shall be provided to ensure adequate lubrication (oil, grease, coolant, etc.) for all moving parts in accordance with manufacturer requirements. Each lubricant reservoir shall have at least a 24-hour capacity. All oil holes, grease fittings, and filler caps shall be easily accessible.

3.1.11 Replacement Parts. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Replacement parts are those that are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts.

3.2 Construction. The machine shall be constructed of parts that are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.2.1 Castings and Forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Fastening Devices. All screws, pins, bolts, and other fasteners shall be installed to prevent unintentional loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.2.3 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28B and the applicable "Detailed Standard" section referenced therein.

3.2.4 Gears. All gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in drive trains shall be hardened and ground steel.

3.2.5 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. The condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.6 Welding, Brazing, or Soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.2.7 Painting. The machine shall be painted in accordance with the manufacturer's commercial practice. All paint shall be of the lead-free type.

3.2.8 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this purchase description.

3.3 Components. The machine shall consist of not less than the following components:

3.3.1 Blasting Enclosure. The PMB booth enclosure shall be constructed as a single, unified structure that includes a retractable partition. This partition shall allow the booth to be divided into two independent workspaces or collapsed to utilize the entire booth as a single operational area. The enclosure shall be dust-tight, effectively containing all blasting media and debris within the booth and preventing any leakage into the surrounding environment. The exterior of the PMB booth shall be fabricated from heavy-duty steel panels, ensuring structural integrity and resistance to the abrasive blasting process. The enclosure, including walls, ceiling, and doors, shall be treated with a protective finish to prevent corrosion and extend durability. The booth shall be completely dust-tight, with all access points and the partition fitted with heavy-duty seals to prevent the escape of dust, debris, and blasting media. The interior walls of the booth shall be lined with a 1/8-inch minimum thick rubber liner to absorb the impact of the blasting media, reduce noise levels, and protect the structural components of the booth from wear and tear.

3.3.1.1 Work Loading Door. The PMB booth enclosure shall have two (2) each work loading doors, one (1) on either end of the booth located closest to the partition. The work loading doors shall be designed in accordance with Table I and be located on the front of the blast booth. Work loading doors shall be provided with an observation window in accordance with manufacturer's standard commercial practice. The work loading doors shall be equipped with leak proof seals on the perimeter to ensure a proper seal and containment of media. The work loading doors shall not obstruct the operation of the personnel doors.

3.3.1.2 Personnel Door. The PMB booth enclosure shall have two (2) each personnel doors, one (1) on either end of the PMB booth, in accordance with Table I. Personnel doors shall be provided with an observation window in accordance with manufacturer's standard commercial practice. Personnel doors shall be located on the front of the PMB booth enclosure, on opposing ends. Personnel doors shall be equipped with leak proof seals on the perimeter to ensure a proper seal and containment of media. The personnel doors shall not obstruct the operation of the main work loading doors.

3.3.2 Differential Pressure Sensor. The design shall include provisions to maintain a negative air pressure of 1/4-inch water gauge (nominal) in the blasting enclosure, with the use of differential pressure sensors, with gauge readouts, across the filter section.

3.3.3 Blasting Equipment. The blasting equipment shall be configured to accommodate one blaster on each side of the enclosure. This setup allows for single-operator use when the enclosure is partitioned, as well as dual-operator use for larger tasks when the partition is removed. The equipment shall be designed for plastic media blasting, with adjustments available to accommodate media sizes of 850-1000 microns, 600-850 microns, and 425-600 microns. The blasting equipment shall be a self-contained closed loop unit.

3.3.3.1 Media Storage Hopper. The media storage hopper shall be designed to allow the full contents of the container to flow freely without the occurrence of media tunneling. The system shall support continuous blasting operations until the hopper is emptied. The media storage hopper shall be equipped with two (2) each butterfly discharge valves, each with a 4-inch diameter, located at the bottom of the hopper. These valves shall be positioned directly above the pressure vessels, allowing them to be opened as necessary to refill the pressure vessels based on operational needs.

3.3.3.2 Pressure Vessel. The contractor shall install four (4) each pressure vessels, two (2) on each side of the PMB booth. The pressure vessel shall conform to ASME BPVC SEC VIII-I and shall be registered with the National Board of Boiler and Pressure Vessel Inspectors for design pressures to 100 psig. The pressure vessel shall be equipped with pressure relief valves set at 90 psig. The pressure vessel shall each have a minimum capacity of 6 cubic feet. All interior surfaces shall be cleaned to remove mill scale and lined or coated to prevent subsequent corrosion. The pressure vessels shall each be fitted with a hand hole or clean out port suitably sized to facilitate inspection and cleaning. Media shall feed automatically into the pressure vessels via a pop-up valve that opens when blasting ceases.

3.3.3.3 Air Pressure Controls/Gauges. Controls shall indicate pressure vessel pressure in psi in increments of 1-psi with an accuracy of +/- 1-psi. Gauges indicating pressure with an accuracy of +/- 1-psi shall be provided for the pressure vessels. Gauge dial faces shall be 3.50-inches in diameter minimum and shall have a range of approximately twice the working pressure.

3.3.3.4 Media Flow Valve. The media flow valve shall be designed so that the manually adjustable metering function and the pneumatically operated valve actuation function are accomplished separately. The plastic media shall be passed from the pressure vessel through a media and air mixing valve to deliver from 0-100% media flow, with +/- 1% through the entire range. The valve shall be mounted at 45 degrees to the compressed air piping and include a clean-out access plate above the flow control area. The metering section of the valve shall incorporate a stainless steel metering plate. The metering valve shall allow fine adjustment from no-flow to fully open.

3.3.3.5 Blast Hose/Nozzle. The blasting system shall include a high-production nozzle with an inner diameter (ID) of 0.5-inch, quick coupling nozzle holder, deadman switch with safety locks, and blast hose ID of 1.125-inch to 1.250-inch. The length of the blast hose shall be no greater than the length of the PMB booth. The nozzle shall be made of boron carbide and connected to the blast hose using an externally mounted nozzle holder made of nylon. The deadman switch shall meet OSHA standards and be securely fastened to the blast hose. Operation shall be pneumatic. Blasting equipment shall be

designed such that nozzle pressure is measured by a hypodermic needle type pressure gage located just behind the nozzle. Control lines for the blast hoses shall be secured to the blast hose and have quick-disconnect fittings allowing control lines to be quickly coupled whenever the blast hose is assembled.

3.3.4 Pneumatic Recovery System. The full floor pneumatic recovery system and all components shall conform to ISO 4414. The system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls.

3.3.4.1 Media Reclaim Equipment. The reclaim system shall receive blasting debris from the media recovery system, remove contaminants, and return the usable media to the blast machine. The system shall consist of a cyclone separator, a rotary airlock, a magnetic particle separator, a specific gravity separator, and a screen separator. The PMB equipment shall be capable of maintaining acceptable levels of media size per MIL-P-85891A and cleanliness of 200 parts per million or less of heavy particle contamination. The system shall be initially set up to process Plastic Type V, Size 30/40. The pneumatic recovery system shall effectively remove all media that falls through the waffle flooring. Provisions to prevent clogging of media return sections shall be incorporated.

3.3.5 Dust Control System. The design shall be in accordance with 29 CFR 1910.94(a). System controls shall be included to ensure the ventilation system continues to run for a minimum of one minute after cessation of blasting to provide clearance of dust laden air within the enclosure. The contractor shall ensure that the system is balanced so that the airflow is laminar throughout the booth.

3.3.5.1 Dust Collector. The dust collectors shall be sized to accommodate the dust laden air streams generated in the enclosure by the blasting operation and the reclaim system. Explosion venting shall be provided where contaminated air, laden with dust or particles, enters the dust collector compartment in accordance with NFPA 68. The dust collector shall be of the pulsejet, cartridge type. Cartridge cleaning shall be automatic and shall include a 0-3 inch water column photohelic type pressure gage to initiate cartridge cleaning. The control system shall initiate the cleaning cycle and be adjustable for continuous cleaning capability. The system shall include a manual override of the pressure differential system when manual cleaning is desired. The dust collector system shall be capable of removing 99.97% of the particles 0.3 micron or greater in size from re-circulation systems. The dust collector shall be fully serviceable from the accessible sides, and the collected material shall be batch-discharged into 55-gallon drums for disposal. The connection from the dust collector hopper to the waste drum shall be dust-tight.

3.3.5.2 Dust Monitor. The dust monitor equipment shall be capable of continuously monitoring explosive conditions of air in the PMB enclosure.

3.3.6 Waste Discharge. The reclaim system waste, with the exception of the dust laden air stream, shall discharge into 55-gallon steel drums via dust tight attachments. Dust laden airstream shall discharge to a dust collector.

3.3.7 Breathing Air Systems. FRCSW shall provide two (2) each, panel mounted Grade-D breathing air filtration systems with carbon monoxide (CO) monitor. One (1) breathing air system for each side of the booth.

3.3.8 Breathing Air Drops. The contractor shall install four (4) each breathing air supply line ports, two (2) ports on each side of the PMB booth. The ports shall be compatible with the Dixon 2FM2-B Industrial Interchange fittings.

3.3.9 Carbon Monoxide Alarm with Strobe. The contractor shall install two (2) each carbon monoxide audible and visual alarms, one (1) on each side of the PMB booth. These alarms shall be provided by FRCSW. The contractor shall install the carbon monoxide alarms in locations that are most practical for the operator inside the booth, ensuring that the alarms are easily visible and audible to provide immediate warning in the event of a carbon monoxide detection.

3.3.10 Compressed Air Drop. The contractor shall install two (2) each 1-inch air piping, one (1) on each side of the PMB booth. The contractor shall install four (4) each dual ¼-inch and ½-inch outlets, two (2) on each side of the PMB booth. Each shall be located by the access door for each half of the booth. All compressed air used by the system shall be obtained by tying into a single point of connection. The government shall provide this point of connection. The contractor is responsible for connecting the equipment to this point.

3.3.11 Electrical System. The machine shall conform to NFPA 79. The existing source power available at the facility is 480 Volt, 3 Phase, 60 Hertz (Hz). The machine's electrical system shall be tolerant of fluctuation of ± 10 percent. The electrical system shall be complete, including any electrical transformer(s) required to modify the existing source voltage to the proper operating voltage of the equipment. A properly rated, and fused, single disconnect device shall be utilized on the machine with means of lockout in the off position only.

3.3.11.1 Electrical Enclosures. Construction of the electrical enclosures shall conform to NEMA ICS 6 standards. The enclosure shall be a NEMA 250 type 12 and shall provide for ventilation of components and shall be designed for an indoor non-hazardous location. All electrical components shall conform to applicable NEMA ICS 1 standards. The machine shall be provided with interlocks on its electronic cabinet that shall instantly remove power from the equipment when the door is opened in accordance with NFPA 79.

3.3.11.2 Arc Flash Protection. The machine enclosures shall be marked for arc flash in accordance with NFPA 79 Paragraph 16.2.3 "Safety Signs for Electrical Enclosures" for arc flash safety.

3.3.11.3 Solid-State Components. Electrical devices used to power electrical servo and spindle motors of control voltage delivered to solid-state electrical components shall be solid-state devices. Such devices shall be equipped to handle the electrical load imposed by operation of the machine to its maximum potential. The use of selenium in solid-state components shall be restricted to those devices that protect other solid-state components from voltage surge.

3.3.12 Interlocks. Electrical interlocks shall be provided so that blasting shall not commence; or if currently operating, shall cease to operate if a work loading or personnel door is open or blast room ventilation not operating.

3.3.13 Control Panels. All electrical functions of the system, except the blasting pressure controls, shall be controlled from a single NEMA enclosure. The tripping of any interlock shall be indicated on this main control panel.

3.3.14 Motors. Motors shall be rated for continuous duty. Alternating-current (AC) motors shall be designed to operate on 60-Hz and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall meet the requirements of NEMA MG-1. Each motor shall have an identification plate in accordance with paragraph 3.6 to identify manufacturer, model number, serial number, voltage, amperage, horsepower, phase, frequency, and frame number. Motor horsepower shall be the manufacturer's standard or standard optional horsepower as indicated on the manufacturer's design and construction drawing(s).

3.3.15 Work Light. Built-in work lighting shall be provided by the contractor. The light shall comply with NFPA 79 requirements, have a protective shield, and provide maximum light to the work area. The work lights shall be installed on the ceiling of the PMB booth. The number and distribution of the lights should be sufficient to provide even illumination across the entire work area, minimizing shadows and ensuring clear visibility of all surfaces. The lights shall be securely mounted to the ceiling, with consideration given to the height and layout of the booth to optimize light coverage. The mounting fixtures should be durable and resistant to vibration and dust generated during blasting operations.

3.3.16 Exhaust Ducting. The contractor shall remove all existing exhaust ducting and ensure all openings are properly sealed and capped to prevent environmental intrusion.

3.3.17 Facility Connections. Electrical connections made to the facility electrical system shall conform to NFPA 70.

3.3.18 Hour Meter. The blast machine shall be equipped with an hour meter to indicate recovery system operation and for determining maintenance schedules. The meter shall be located on the main equipment control panel. The meter shall be of the non-resetting type and shall have a range of 0 to 99,999 hours in increments of 1/10-hour. The meter shall automatically reset to 0 hours after 100,000 hours have passed. The meter shall be designed to prevent the entrance of dust and moisture.

3.4 Size and Capacity. The size and capacity of the machine shall be not less than specified in Table I.

Materials to be blasted	Steel, Aluminum, and Composites
Part size minimum	1ftx1ftx1ft
Part size Maximum	16ftx4ftx2ft
Blast Booth size Minimum	35ft wide x 12 high x12.5 ft deep
Blast Booth size Maximum	To fit with in total floor space
Total floor space including ancillary equipment	49ft 2in Long x 24ft 7in Wide
Floor to I Beam	25 Ft
Work Loading Doors	10ft 10in Long x 24ft 7in Wide
Personnel Doors	36in Wide x 84in Tall
Accordion Partition	12ft Tall x 12ft 6in Long
The blast booth footprint has a total of 3 depressions that make up the total foundation area:	(1) 13.5ft Length x 34ft Width x 8.5in Depth (2) 12ft Length x 34ft' Width x 14in Depth (3) 3ft-3in Length x 40ft Width x 8.5in Depth

3.5 Standard Tooling and Equipment. All standard tooling and equipment required for proper operation of the machine (such as special tools needed for maintenance) shall be furnished. A list of these items shall be furnished at time of proposal.

3.6 Marking on Instruments, Control Panels, and Plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion resistant.

3.6.1 Lubrication Plate. A lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

- Points of Lubricant Application
- Servicing Interval
- Type of Lubricant
- Viscosity
- Military or Federal Specification for Each Lubricant (If available)
- Size and Type of Each Lubricant Filter

3.6.2 Nameplate. A nameplate shall be permanently and securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

- Nomenclature
- Manufacturer's Name
- Manufacturer's Model Designation
- Manufacturer's Serial Number

Power Input (Volts, Total Amps, Phase, Frequency)
Contract Number
Date of Manufacture

3.6.3 Identification Marking of Military Property. An Item Unique Identification Marking (IUID) shall be provided in accordance with MIL-STD-130N and all applicable documents within the standard with Machine Readable Information (MRI) for item identification marking and automatic data capture. The application of Human Readable Information (HRI) shall be used in combination with MRI and free text. At a minimum this tag shall contain the information listed in paragraph 3.7.2 Nameplate.

3.6.4 Technical Data. Three (3) hard copies and one (1) electronic copy of technical data shall be provided with each blast system. All technical data shall be legible, written in English, and include the following elements:

- a. All engineering drawings shall comply with ASME Y14.100 standards. The technical data shall include technical specifications, instructions for the operation and maintenance of the machine, and information on consumables and expendables, such as oils, fluids, grease, and filters. Additionally, a recommended spare parts list shall be provided.
- b. The contractor shall supply factory service and operator manuals specific to the type and model of the machine being purchased.
- c. Diagnostic procedures for isolating and correcting faults shall be included. The service manuals shall contain procedures for recalibrating and performing operational checks to ensure the proper operation of any replaced parts.
- d. All software manuals, along with the necessary registration keys/codes and original software installation disks, shall be provided.
- e. Two (2) copies of the manufacturer's standard operational and performance test procedures shall be submitted thirty (30) calendar days After Contract Award (ACA) for Government review and approval. The approved test procedures shall become the property of the Government.

3.6.5 Plan of Action and Milestones (POA&M). The contractor shall provide a POA&M to identify contractor performance dates of all major events from contract award to completion of all contract requirements. The initial POA&M shall be delivered thirty (30) calendar days After Contract Award (ACA). The milestones shall be prepared using a recognized project management technique (preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M shall be approved in advance by the Government. The POA&M shall detail such items as engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

3.7 Demolition and Removal of Existing PMB Booth. The contractor shall purge the existing PMB booth system of all fluids and blast media. FRCSW shall provide metal drums for the proper disposal of blast media. The contractor shall be responsible for separating the materials and labeling the drums according to their contents. The contractor shall notify the government at least fourteen (14) workdays prior to the start of the removal process. The removal of utilities shall include, but is not limited to, associated components such as piping, conduit, and wiring. The contractor shall transport the removed equipment to designated bins outside of the Blast Shop. Upon removal of all materials and equipment, the building structure and any affected areas shall be restored, patched, and sealed to prevent environmental intrusion. FRCSW shall be responsible for the disposal of all waste.

3.8 Foundation. The contractor shall modify the existing foundation as needed to accommodate the footprint of the new PMB blast booth.

3.9 Installation. The contractor shall be responsible for all equipment (forklifts, man lifts, etc.) and qualified personnel required for unloading, rigging, installation, foundation work, anchoring, connecting machine to utilities, and leveling the machine in place. The contractor shall provide the necessary anchor bolts, leveling screws and related hardware to secure the machine components to a foundation. The contractor shall also be responsible for the Installation Responsibilities sheet. The contractor shall be responsible for reading, understanding, and complying with FRCSW facility requirements.

3.9.1 Workdays and Hours. Services provided by the contractor or sub-contractors shall be performed IAW FRCSW normal workdays Monday through Friday, during the hours of 6:00am and 2:30pm, excluding Federal holidays, furloughs, scheduled shutdown between Christmas and New Year's, and base closures unless previously arranged through the contracting officer.

3.10 Training. After satisfactory completion of acceptance testing of the system, the services of a qualified representative(s) shall be provided for specialized training to familiarize Government personnel with the equipment and to help ensure reliable performance and maximum service life, during normal usage. The course shall utilize as-built drawings and manuals provided under this contract and shall include both classroom and hands-on portions. All printed training aids shall be supplied for the course, be in the English language, and become the property of the Government.

3.10.1 Operator Training. Operator training shall be provided at FRCSW. Training shall be provided for a minimum of five (5) operators. Training shall not be less than two (2) consecutive eight (8) hour workdays, IAW days and hours listed in 3.9.1. Training shall include preparation of equipment for safe operation and proper isolation of equipment procedures. The operator shall be provided instruction for the care, use and operation of all attachments and accessories. Training manuals shall be provided for each student.

3.10.2 Maintenance Training. Mechanical and Electronic/Electrical maintenance training shall be provided for five (5) engineer/maintenance personnel. Training shall be performed at the FRCSW and shall not be less than three (3) consecutive eight (8) hour workdays each, IAW days and hours. Training manuals shall be provided for each student. The maintenance course shall cover at a minimum:

- a. Review of systems drawings and schematics
- b. Component location and function
- a. Troubleshooting procedures and techniques
- c. Repair procedures including disassembly and assembly
- b. System maintenance and servicing
- d. Setup, adjustments, and calibration (how, when, and where)
- c. Preventative maintenance procedures
- h. Procedures for proper isolation.

4.0 QUALITY ASSURANCE.

4.1 Responsibility for Inspection. The contractor shall be responsible for the performance of all inspection requirements specified herein and in accordance with the attached Quality Assurance Provision (QAP). The Government reserves the right to witness any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for Compliance. All items shall meet all requirements of sections 3 and 4.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

- a. Final Inspection/Acceptance test (Destination) (see 4.3).

4.3 Final Inspection/Acceptance Test (Destination). Final Inspection and acceptance test shall be performed on the machine to ensure conformance with this purchase description. The acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.4 and all tests in 4.5. The machine shall pass the examination and all tests to be accepted.

4.4 Examination. The equipment shall be examined for design, dimensions, construction, materials, components, electrical equipment and workmanship to determine compliance with the requirements of this specification.

4.5 Tests. Unless otherwise specified within this document, all programs, instruments, materials, material handling equipment, qualified personnel, and tools required to perform and evaluate the tests shall be furnished by the contractor. All measurement tools and indicating devices shall be calibrated in accordance with the attached QAP. If a malfunction occurs during the test, the test shall be restarted each time until all tests in 4.5

are completed without a malfunction. Three (3) consecutive failures without completion of the test may be considered cause for rejection of the system. For the purpose of this test, a “failure” is defined as any equipment malfunction which requires remedial action to restore the system to full operation in accordance with this specification.

4.5.1 Operational Test. The contractor shall provide a custom PMB system acceptance check list that ensures all equipment and accessories are fully functional and operational in order to fulfill a successful plastic media blasting operation. The complete system shall be connected to the appropriate utility sources. The system shall then be prepared for corrosion removal and surface cleaning in accordance with the manufacturer's published operating instructions. The system shall be tested under operating conditions for not less than 45 minutes. Operational performance of all sub-systems and components which make up the entire blast booth, dust collector, air handling units shall be verified and recorded to ensure mechanical functionality.

4.5.2 Performance Test. The system shall be tested for its designed surface preparation capabilities following the parameters specified in MIL-P-85891A. The Government shall provide blast media and two test pieces at destination inspection. Preparation of the metal patch specimen, including all pre and post preparation requirements shall be the responsibility of the contractor with assistance from FRCSW.

4.5.3 Leak Test. The contractor shall perform a comprehensive leak test on the fully assembled PMB booth to ensure system integrity and compliance with operational standards. The leak test shall be conducted by pressurizing the booth's air handling, dust collection, and pneumatic systems to their specified operational pressures. Any drop in pressure beyond acceptable limits or visible evidence of air or media leakage shall be immediately addressed. Upon completion, the contractor shall provide a report detailing the test results, including identified leaks and corrective actions taken. All repairs shall be verified with a follow-up leak test to ensure system integrity before final acceptance.

4.5.4 Deadman Switch Operation. The deadman switch shall be tested to ensure proper functionality. Upon depression of the switch, media flow shall reach steady state conditions within 10 seconds. Upon release of the switch, media flow shall cease completely within 5 seconds.

4.5.5 Interlocks Testing. The system shall be tested to demonstrate the conformance of each individual interlock in Section 3.3.12.

4.5.6 Audible Noise Level Test. The equipment shall be tested for compliance with paragraph 3.1.4. “Audible Noise Level”.

1. INSTALLATION RESPONSIBILITIES

Government	Contractor	
☐	☒	a. Provide machine foundation in accordance with the Manufacturer's design requirements
☐	☒	b. Furnish labor and material handling equipment for off-loading and placing item on foundation
☐	☒	c. Provide and install anchor bolts and nuts
☐	☒	d. Set and rough level the machine on its foundation
☐	☒	e. Connect machine to the provided utilities hook-up
☐	☒	f. Provide all necessary tools, gages, and instrumentation necessary to perform the required tests
☐	☒	g. Provide and charge all systems with fluids in accordance with manufacturer's instructions
☐	☒	h. Modify existing foundation to accommodate new PMB blast booth
☐	☒	i. Material handling, Rigging equipment and Qualified Personnel for performance testing
☒	☐	j. Material for performance testing at Government facility

GOVERNMENT RESPONSIBILITIES

- ☒ Provide utilities hook-up within 20 feet of the machine.